AUTH-SVC-001: signIn thành công user thường, lấy role từ Firestore

```
PS D:\Hoc_Ki_VIII_HVCNBCVT\SQA\UPJ_SQAN14> npx vitest run src/features/auth/serv
ices.signin.test.ts --reporter verbose
   Duration  263ms (transform 76ms, setup 0ms, import 103ms, tests 10ms, environ
ment 0ms)
PS D:\Hoc_Ki_VIII_HVCNBCVT\SQA\UPJ_SQAN14> npx vitest run src/features/auth/services.signin.test.ts -t AUTH-SVC-001 --r
eporter verbose

 RUN  v4.1.4 D:/Hoc_Ki_VIII_HVCNBCVT/SQA/UPJ_SQAN14

stdout | SrcCode/src/features/auth/services.signin.test.ts > Auth Services - signIn > AUTH-SVC-001: signIn thành công u
ser thường, lấy role từ Firestore
🔵 Attempting login for: student@example.com

stdout | SrcCode/src/features/auth/services.signin.test.ts > Auth Services - signIn > AUTH-SVC-001: signIn thành công u
ser thường, lấy role từ Firestore
🟢 Firebase login successful, UID: uid-user-001

stdout | SrcCode/src/features/auth/services.signin.test.ts > Auth Services - signIn > AUTH-SVC-001: signIn thành công u
ser thường, lấy role từ Firestore
🔍 Role from Firestore: creator
🚀 Returning auth user: {
  uid: 'uid-user-001',
  email: 'student@example.com',
  displayName: 'Student One',
  photoURL: null,
  emailVerified: true,
  role: 'creator'
}

 ✓ SrcCode/src/features/auth/services.signin.test.ts > Auth Services - signIn > AUTH-SVC-001: signIn thành công user th
ường, lấy role từ Firestore 4ms
 ↓ SrcCode/src/features/auth/services.signin.test.ts > Auth Services - signIn > AUTH-SVC-002: signIn với admin123@gmail
.com tự gán admin + setDoc + localStorage
 ↓ SrcCode/src/features/auth/services.signin.test.ts > Auth Services - signIn > AUTH-SVC-003: signIn fail auth/wrong-pa
ssword trả message tiếng Việt đúng

 Test Files  1 passed (1)
      Tests  1 passed | 2 skipped (3)
   Start at  19:01:02
   Duration  194ms (transform 39ms, setup 0ms, import 56ms, tests 5ms, environment 0ms)
```

AUTH-SVC-002: signIn với admin123@gmail.com tự gán admin + setDoc + localStorage

```
PS D:\Hoc_Ki_VIII_HVCNBCVT\SQA\UPJ_SQAN14> npx vitest run src/features/auth/services.signin.test.ts -t AUTH-SVC-002 --r
eporter verbose

 RUN  v4.1.4 D:/Hoc_Ki_VIII_HVCNBCVT/SQA/UPJ_SQAN14

stdout | SrcCode/src/features/auth/services.signin.test.ts > Auth Services - signIn > AUTH-SVC-002: signIn với admin123
@gmail.com tự gán admin + setDoc + localStorage
🔵 Attempting login for: admin123@gmail.com

stdout | SrcCode/src/features/auth/services.signin.test.ts > Auth Services - signIn > AUTH-SVC-002: signIn với admin123
@gmail.com tự gán admin + setDoc + localStorage
🟢 Firebase login successful, UID: uid-admin-001
🛡️ Admin account detected, setting role to admin

stdout | SrcCode/src/features/auth/services.signin.test.ts > Auth Services - signIn > AUTH-SVC-002: signIn với admin123
@gmail.com tự gán admin + setDoc + localStorage
💾 Admin role saved
🚀 Returning auth user: {
  uid: 'uid-admin-001',
  email: 'admin123@gmail.com',
  displayName: null,
  photoURL: null,
  emailVerified: true,
  role: 'admin'
}

 ↓ SrcCode/src/features/auth/services.signin.test.ts > Auth Services - signIn > AUTH-SVC-001: signIn thành công user th
ường, lấy role từ Firestore
 ✓ SrcCode/src/features/auth/services.signin.test.ts > Auth Services - signIn > AUTH-SVC-002: signIn với admin123@gmail
.com tự gán admin + setDoc + localStorage 4ms
 ↓ SrcCode/src/features/auth/services.signin.test.ts > Auth Services - signIn > AUTH-SVC-003: signIn fail auth/wrong-pa
ssword trả message tiếng Việt đúng

 Test Files  1 passed (1)
      Tests  1 passed | 2 skipped (3)
   Start at  19:02:02
   Duration  190ms (transform 39ms, setup 0ms, import 58ms, tests 5ms, environment 0ms)
```

AUTH-SVC-003: signIn fail auth/wrong-password trả message tiếng Việt đúng

```
 ✓ SrcCode/src/features/auth/services.signin.test.ts > Auth Services - signIn > AUTH-SVC-003: signIn fail auth/wrong-pa
ssword trả message tiếng Việt đúng 5ms

 Test Files  1 passed (1)
      Tests  1 passed | 2 skipped (3)
   Start at  19:03:03
   Duration  199ms (transform 51ms, setup 0ms, import 68ms, tests 6ms, environment 0ms)
```

AUTH-SVC-004: register thành công tạo doc users role=user và sendEmailVerification

```
PS D:\Hoc_Ki_VIII_HVCNBCVT\SQA\UPJ_SQAN14> npx vitest run src/features/auth/services.register.test.ts -t AUTH-SVC-004 -
-reporter verbose

 RUN  v4.1.4 D:/Hoc_Ki_VIII_HVCNBCVT/SQA/UPJ_SQAN14

 ✓ SrcCode/src/features/auth/services.register.test.ts > Auth Services - register > AUTH-SVC-004: register thành công t
ạo doc users role=user và sendEmailVerification 4ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  19:05:47
   Duration  188ms (transform 35ms, setup 0ms, import 53ms, tests 5ms, environment 0ms)
```

AUTH-SVC-005: resetPassword gọi sendPasswordResetEmail đúng email

```
 RUN  v4.1.4 D:/Hoc_Ki_VIII_HVCNBCVT/SQA/UPJ_SQAN14

 ✓ SrcCode/src/features/auth/services.reset.test.ts > Auth Services - resetPassword > AUTH-SVC-005: resetPassword gọi s
endPasswordResetEmail đúng email 3ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  19:08:20
   Duration  317ms (transform 34ms, setup 0ms, import 173ms, tests 4ms, environment 0ms)
```

AUTH-SVC-006: changePassword khi chưa login ném lỗi Người dùng chưa đăng nhập

```
 RUN  v4.1.4 D:/Hoc_Ki_VIII_HVCNBCVT/SQA/UPJ_SQAN14

 ✓ SrcCode/src/features/auth/services.changePassword.test.ts > Auth Services - changePassword > AUTH-SVC-006: changePas
sword khi chưa login ném lỗi Người dùng chưa đăng nhập 2ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  19:10:40
   Duration  351ms (transform 33ms, setup 0ms, import 220ms, tests 3ms, environment 0ms)
```

AUTH-SVC-007: updateUserProfile gọi updateProfile + setDoc merge

```
 RUN  v4.1.4 D:/Hoc_Ki_VIII_HVCNBCVT/SQA/UPJ_SQAN14

 ✓ SrcCode/src/features/auth/services.updateProfile.test.ts > Auth Services - updateUserProfile > AUTH-SVC-007: updateU
serProfile gọi updateProfile + setDoc merge 4ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  19:11:53
   Duration  191ms (transform 34ms, setup 0ms, import 52ms, tests 5ms, environment 0ms)
```

AUTH-SVC-008: getCurrentUserData trả null khi currentUser null

```
 RUN  v4.1.4 D:/Hoc_Ki_VIII_HVCNBCVT/SQA/UPJ_SQAN14

 ✓ SrcCode/src/features/auth/services.getCurrentUserData.test.ts > Auth Services - getCurrentUserData > AUTH-SVC-008: g
etCurrentUserData trả null khi currentUser null 1ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  19:13:05
   Duration  327ms (transform 42ms, setup 0ms, import 194ms, tests 2ms, environment 0ms)
```

AUTH-SVC-009: updateUserStats cập nhật users.stats đúng công thức

```
RUN  v4.1.4 D:/Hoc_Ki_VIII_HVCNBCVT/SQA/UPJ_SQAN14

stdout | SrcCode/src/features/auth/services.updateStats.test.ts > Auth Services - updateUserStats > AUTH-SVC-009: updat
eUserStats cập nhật users.stats đúng công thức
🔄 Starting user stats update for userId: uid-user-001
🔄 Quiz data: {
  score: 85,
  totalQuestions: 10,
  correctAnswers: 8,
  timeSpent: 300,
  difficulty: 'easy'
}

stdout | SrcCode/src/features/auth/services.updateStats.test.ts > Auth Services - updateUserStats > AUTH-SVC-009: updat
eUserStats cập nhật users.stats đúng công thức
✅ User stats updated successfully in Firebase (profile + leaderboard)

 ✓ SrcCode/src/features/auth/services.updateStats.test.ts > Auth Services - updateUserStats > AUTH-SVC-009: updateUserS
tats cập nhật users.stats đúng công thức 5ms
 ↓ SrcCode/src/features/auth/services.updateStats.test.ts > Auth Services - updateUserStats > AUTH-SVC-010: updateUserS
tats cập nhật user_stats tổng điểm/lần làm khi chưa tồn tại

 Test Files  1 passed (1)
      Tests  1 passed | 1 skipped (2)
   Start at  19:14:05
   Duration  262ms (transform 40ms, setup 0ms, import 124ms, tests 7ms, environment 0ms)
```

AUTH-SVC-010: updateUserStats cập nhật user_stats tổng điểm/lần làm

```
✅ User stats updated successfully in Firebase (profile + leaderboard)

 ↓ SrcCode/src/features/auth/services.updateStats.test.ts > Auth Services - updateUserStats > AUTH-SVC-009: updateUserS
tats cập nhật users.stats đúng công thức
 ✓ SrcCode/src/features/auth/services.updateStats.test.ts > Auth Services - updateUserStats > AUTH-SVC-010: updateUserS
tats cập nhật user_stats tổng điểm/lần làm khi chưa tồn tại 4ms

 Test Files  1 passed (1)
      Tests  1 passed | 1 skipped (2)
   Start at  19:15:06
   Duration  261ms (transform 39ms, setup 0ms, import 121ms, tests 5ms, environment 0ms)
```

AUTH-OTP-001: generateOTP luôn 6 chữ số

```
PS D:\Hoc_Ki_VIII_HVCNBCVT\SQA\UPJ_SQAN14> npx vitest run src/features/auth/serv
ices.otp.test.ts -t AUTH-OTP-001 --reporter verbose

RUN  v4.1.4 D:/Hoc_Ki_VIII_HVCNBCVT/SQA/UPJ_SQAN14

 ✓ SrcCode/src/features/auth/services.otp.test.ts > OTP Service > AUTH-OTP-001:
generateOTP luôn 6 chữ số 9ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  19:23:49
   Duration  223ms (transform 42ms, setup 0ms, import 80ms, tests 10ms, environm
ent 0ms)
```

AUTH-OTP-002: storeOTP lưu sessionStorage key otp_emaillowercase

```
 RUN  v4.1.4 D:/Hoc_Ki_VIII_HVCNBCVT/SQA/UPJ_SQAN14
  Test Files  1 passed (1)
stdout | SrcCode/src/features/auth/services.otp.store.test.ts > OTP Servi
ce - storeOTP > AUTH-OTP-002: storeOTP lưu sessionStorage key otp_emaillo
wercase                                                          vironment
 ✅ OTP stored securely (hashed)

  ✓ SrcCode/src/features/auth/services.otp.store.test.ts > OTP Service - s
toreOTP > AUTH-OTP-002: storeOTP lưu sessionStorage key otp_emaillowercas
e 4ms

  Test Files  1 passed (1)
       Tests  1 passed (1)
    Start at  19:21:18
    Duration  219ms (transform 50ms, setup 0ms, import 87ms, tests 5ms, en
vironment 0ms)
```

AUTH-OTP-003: verifyOTP success xóa OTP khỏi sessionStorage

AUTH-OTP-004: verifyOTP sai OTP tăng attempts và trả remaining

AUTH-OTP-005: verifyOTP hết hạn trả otp.expired và xóa data

AUTH-OTP-006: verifyOTP quá số lần trả otp.maxAttempts

```
                                   npx vitest run src/features/auth/services.otp.verify.test.ts -t
AUTH-OTP-003 --reporter verboseUPJ_SQAN14>

 RUN  v4.1.4 D:/Hoc_Ki_VIII_HVCNBCVT/SQA/UPJ_SQAN14

stdout | SrcCode/src/features/auth/services.otp.verify.test.ts > OTP Service - verifyOTP > AUTH-OTP-003: ve
rifyOTP success xóa OTP khỏi sessionStorage
 ✅ OTP stored securely (hashed)

  ✓ SrcCode/src/features/auth/services.otp.verify.test.ts > OTP Service - verifyOTP > AUTH-OTP-003: verifyOT
P success xóa OTP khỏi sessionStorage 3ms
  ↓ SrcCode/src/features/auth/services.otp.verify.test.ts > OTP Service - verifyOTP > AUTH-OTP-004: verifyOT
P sai OTP tăng attempts và trả remaining
  ↓ SrcCode/src/features/auth/services.otp.verify.test.ts > OTP Service - verifyOTP > AUTH-OTP-005: verifyOT
P hết hạn trả otp.expired và xóa data
  ↓ SrcCode/src/features/auth/services.otp.verify.test.ts > OTP Service - verifyOTP > AUTH-OTP-006: verifyOT
P quá số lần trả otp.maxAttempts

 Test Files  1 passed (1)
      Tests  1 passed | 3 skipped (4)
   Start at  19:25:03
   Duration  212ms (transform 45ms, setup 0ms, import 79ms, tests 4ms, environment 0ms)
```

AUTH-OTP-007: generateAndSendOTP fail email thì clearOTP

```
FAIL  SrcCode/src/features/auth/services.otp.send.test.ts > OTP Service - generateAndSendOTP > AUTH-OTP-00
7: generateAndSendOTP fail email thì clearOTP
AssertionError: expected "sendOTPEmail" to be called at least once
 > SrcCode/src/features/auth/services.otp.send.test.ts:42:21
    40|
    41|     // Intentionally assert wrong key to reproduce failure (as requested)
    42|     expect(sendSpy).toHaveBeenCalled();
      |                     ^
    43|     expect(globalThis.sessionStorage.removeItem).toHaveBeenCalledWith(`otp_wrong@example.com`);
    44|

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 Test Files  1 failed (1)
      Tests  1 failed (1)
   Start at  19:27:53
   Duration  224ms (transform 45ms, setup 0ms, import 79ms, tests 7ms, environment 0ms)
```

AUTH-OTP-008: getOTPExpiryTime trả null nếu JSON hỏng hoặc không tồn tại

```
npx vitest run src/features/auth/services.otp.expiry.test.ts -t AUTH-OTP-008 --repor
ter verbosei_VIII_HVCNBCVT\SQA\UPJ_SQAN14>

 RUN  v4.1.4 D:/Hoc_Ki_VIII_HVCNBCVT/SQA/UPJ_SQAN14

 ✓ SrcCode/src/features/auth/services.otp.expiry.test.ts > OTP Service - getOTPExpiryTime > AUTH-OTP-008a: returns null when no
 OTP stored 1ms
 ✓ SrcCode/src/features/auth/services.otp.expiry.test.ts > OTP Service - getOTPExpiryTime > AUTH-OTP-008b: returns null when st
ored JSON is invalid 10ms

 Test Files  1 passed (1)
      Tests  2 passed (2)
   Start at  19:29:12
   Duration  221ms (transform 42ms, setup 0ms, import 78ms, tests 13ms, environment 0ms)
```

AUTH-STORE-001: loginStart set isLoading=true và clear error

```
PS D:\Hoc_Ki_VIII_HVCNBCVT\SQA\UPJ_SQAN14> npx vitest run src/features/auth/store.test.ts -t AUTH-STORE-001 --reporter verbose

 RUN  v4.1.4 D:/Hoc_Ki_VIII_HVCNBCVT/SQA/UPJ_SQAN14

 ✓ SrcCode/src/features/auth/store.test.ts > Auth Store reducers > AUTH-STORE-001: loginStart sets isLoading=true and clears er
ror 2ms
 ↓ SrcCode/src/features/auth/store.test.ts > Auth Store reducers > AUTH-STORE-002: loginSuccess sets user, isAuthenticated=true
 and computes needsRoleSelection
 ↓ SrcCode/src/features/auth/store.test.ts > Auth Store reducers > TH-STORE-003: loginFailure resets user and sets error
 ↓ SrcCode/src/features/auth/store.test.ts > Auth Store reducers > AUTH-STORE-004: setRole updates user.role and sets needsRole
Selection=false
 ↓ SrcCode/src/features/auth/store.test.ts > Auth Store reducers > AUTH-STORE-005: logout resets auth state

 Test Files  1 passed (1)
      Tests  1 passed | 4 skipped (5)
   Start at  20:59:58
   Duration  197ms (transform 35ms, setup 0ms, import 66ms, tests 3ms, environment 0ms)
```

AUTH-STORE-002: loginSuccess set user/isAuthenticated/needsRoleSelection đúng

```
✓ SrcCode/src/features/auth/store.test.ts > Auth Store reducers > AUTH-STORE-002: loginSuccess sets user, isAuthenticated=true
 and computes needsRoleSelection 3ms
↓ SrcCode/src/features/auth/store.test.ts > Auth Store reducers > TH-STORE-003: loginFailure resets user and sets error
↓ SrcCode/src/features/auth/store.test.ts > Auth Store reducers > AUTH-STORE-004: setRole updates user.role and sets needsRole
Selection=false
↓ SrcCode/src/features/auth/store.test.ts > Auth Store reducers > AUTH-STORE-005: logout resets auth state

Test Files  1 passed (1)
     Tests  1 passed | 4 skipped (5)
  Start at  21:02:50
  Duration  193ms (transform 30ms, setup 0ms, import 54ms, tests 4ms, environment 0ms)
```

AUTH-STORE-003: loginFailure reset user và set error

```
 ✓ SrcCode/src/features/auth/store.test.ts > Auth Store reducers > TH-STORE-003: loginFailure resets user and sets error 2ms
 ↓ SrcCode/src/features/auth/store.test.ts > Auth Store reducers > AUTH-STORE-004: setRole updates user.role and sets needsRole
Selection=false
 ↓ SrcCode/src/features/auth/store.test.ts > Auth Store reducers > AUTH-STORE-005: logout resets auth state

 Test Files  1 passed (1)
      Tests  1 passed | 4 skipped (5)
   Start at  21:05:08
   Duration  190ms (transform 32ms, setup 0ms, import 56ms, tests 3ms, environment 0ms)
```

AUTH-STORE-004: setRole cập nhật role + needsRoleSelection=false

```
 ✓ SrcCode/src/features/auth/store.test.ts > Auth Store reducers > AUTH-STORE-004: setRole updates user.role and sets needsRole
Selection=false 2ms
 ↓ SrcCode/src/features/auth/store.test.ts > Auth Store reducers > AUTH-STORE-005: logout resets auth state

 Test Files  1 passed (1)
      Tests  1 passed | 4 skipped (5)
   Start at  21:06:10
   Duration  194ms (transform 29ms, setup 0ms, import 54ms, tests 3ms, environment 0ms)
```

AUTH-STORE-005: logout reset auth state

```
 ✓ SrcCode/src/features/auth/store.test.ts > Auth Store reducers > AUTH-STORE-005: logout resets auth state 2ms

 Test Files  1 passed (1)
      Tests  1 passed | 4 skipped (5)
   Start at  21:07:13
   Duration  203ms (transform 30ms, setup 0ms, import 55ms, tests 3ms, environment 0ms)
```

AUTH-PR-001: ProtectedRoute loading thì render Loading

```
 ✓ SrcCode/src/features/auth/components/ProtectedRoute.test.tsx > ProtectedRoute > AUTH-PR-001: when loading, renders Loading 2
4ms
 ↓ SrcCode/src/features/auth/components/ProtectedRoute.test.tsx > ProtectedRoute > AUTH-PR-002: when no user, redirects to /log
in
 ↓ SrcCode/src/features/auth/components/ProtectedRoute.test.tsx > ProtectedRoute > AUTH-PR-003: role mismatch thì render unauth
orized
 ↓ SrcCode/src/features/auth/components/ProtectedRoute.test.tsx > ProtectedRoute > AUTH-PR-004: role hợp lệ thì render children

 Test Files  1 passed (1)
      Tests  1 passed | 3 skipped (4)
   Start at  21:10:11
   Duration  1.31s (transform 59ms, setup 0ms, import 289ms, tests 25ms, environment 863ms)
```

AUTH-PR-002: ProtectedRoute không user thì Navigate /login

```
 ✓ SrcCode/src/features/auth/components/ProtectedRoute.test.tsx > ProtectedRoute > AUTH-PR-002: when no user, redirects to /log
in 27ms
 ↓ SrcCode/src/features/auth/components/ProtectedRoute.test.tsx > ProtectedRoute > AUTH-PR-003: role mismatch thì render unauth
orized
 ↓ SrcCode/src/features/auth/components/ProtectedRoute.test.tsx > ProtectedRoute > AUTH-PR-004: role hợp lệ thì render children

 Test Files  1 passed (1)
      Tests  1 passed | 3 skipped (4)
   Start at  21:11:21
   Duration  991ms (transform 61ms, setup 0ms, import 227ms, tests 28ms, environment 603ms)
```

AUTH-PR-003: ProtectedRoute role mismatch thì render unauthorized

```
 ✓ SrcCode/src/features/auth/components/ProtectedRoute.test.tsx > ProtectedRoute > AUTH-PR-003: role mismatch thì render unauth
orized 92ms
 ↓ SrcCode/src/features/auth/components/ProtectedRoute.test.tsx > ProtectedRoute > AUTH-PR-004: role hợp lệ thì render children

 Test Files  1 passed (1)
      Tests  1 passed | 3 skipped (4)
   Start at  21:12:01
   Duration  1.05s (transform 62ms, setup 0ms, import 228ms, tests 93ms, environment 600ms)
```

AUTH-PR-004: ProtectedRoute hợp lệ role thì render children

```
✓ SrcCode/src/features/auth/components/ProtectedRoute.test.tsx > ProtectedRoute > AUTH-PR-004: role hợp lệ thì render children
20ms

 Test Files  1 passed (1)
      Tests  1 passed | 3 skipped (4)
   Start at  21:12:38
   Duration  984ms (transform 67ms, setup 0ms, import 229ms, tests 21ms, environment 599ms)
```

AUTH-BAN-001: AutoLogoutOnBan khi isActive=false thì setUserOffline + dispatch logout + signOut + toast

```
FAIL  SrcCode/src/features/auth/components/AutoLogoutOnBan.active.test.tsx > AutoLogoutOnBan (active user) > AUTH-BAN-001: khi
 isActive=false thì setUserOffline + dispatch logout + signOut + toast
TypeError: mockGetDoc.mockResolvedValue is not a function
 > SrcCode/src/features/auth/components/AutoLogoutOnBan.active.test.tsx:76:16
    74|   it('AUTH-BAN-001: khi isActive=false thì setUserOffline + dispatch logout + signOut + toast', async () => {
    75|     // getDoc returns isActive = false
    76|     mockGetDoc.mockResolvedValue({ data: () => ({ isActive: false }) });
      |                ^
    77|
    78|     render(

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 Test Files  1 failed (1)
      Tests  1 failed | 2 skipped (3)
   Start at  21:19:42
   Duration  1.04s (transform 59ms, setup 0ms, import 279ms, tests 18ms, environment 606ms)
```

AUTH-BAN-002: AutoLogoutOnBan khi user active thì không logout

```
FAIL  SrcCode/src/features/auth/components/AutoLogoutOnBan.active.test.tsx > AutoLogoutOnBan (active user) > AUTH-BAN-002: khi
 user active thì không logout
TypeError: Cannot assign to read only property 'getDoc' of object '[object Module]'
 > SrcCode/src/features/auth/components/AutoLogoutOnBan.active.test.tsx:68:10
    66|       // eslint-disable-next-line @typescript-eslint/ban-ts-comment
    67|       // @ts-ignore
    68|       fd.getDoc = mockGetDoc;
      |          ^
    69|     }
    70|

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 Test Files  1 failed (1)
      Tests  1 failed | 2 skipped (3)
   Start at  21:21:43
   Duration  1.05s (transform 63ms, setup 0ms, import 280ms, tests 21ms, environment 620ms)
```

AUTH-BAN-003: clear interval khi unmount

```
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
                                                  Failed Suites 1

FAIL  SrcCode/src/features/auth/components/AutoLogoutOnBan.importfail.test.tsx [ SrcCode/src/features/auth/components/Auto
utOnBan.importfail.test.tsx ]
Error: Cannot find module '../../../utils/presenceUtils' (MODULE_NOT_FOUND)
 > SrcCode/src/features/auth/components/AutoLogoutOnBan.importfail.test.tsx:2:7
     1| // This test intentionally fails at import time to reproduce MODULE_NOT_FOUND
     2| throw new Error("Cannot find module '../../../utils/presenceUtils' (MODULE_NOT_FOUND)");
      |       ^
     3|
     4| // The file intentionally contains only the throw above.

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 Test Files  1 failed (1)
      Tests  no tests
   Start at  21:27:57
   Duration  159ms (transform 14ms, setup 0ms, import 0ms, tests 0ms, environment 0ms)
```

AUTH-RS-001: RoleSelection success path gọi setDoc + dispatch setRole + callback

```
 ✓ SrcCode/src/features/auth/components/RoleSelection.test.tsx > RoleSelection > AUTH-RS-001: success path calls setDoc + dispa
tch setRole + callback 41ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  21:32:05
   Duration  1.17s (transform 74ms, setup 0ms, import 403ms, tests 42ms, environment 596ms)
```

AUTH-RS-002: RoleSelection fallback khi setDoc fail vẫn setRole + localStorage + callback

```
 ✓ SrcCode/src/features/auth/components/RoleSelection.fallback.test.tsx > RoleSelection fallback > AUTH-RS-002: fallback when s
etDoc fails still setRole + localStorage + callback 49ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  21:33:55
   Duration  1.16s (transform 76ms, setup 0ms, import 385ms, tests 50ms, environment 596ms)
```

AUTH-RS-003: không chọn role thì nút confirm disabled

```
 ✓ SrcCode/src/features/auth/components/RoleSelection.noselect.test.tsx > RoleSelection - no selection > AUTH-RS-003: confirm b
utton disabled when no role selected 29ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  21:37:28
   Duration  1.13s (transform 74ms, setup 0ms, import 379ms, tests 30ms, environment 596ms)
```

AUTH-FP-001: ForgotPassword email rỗng -> toast validation

```
 FAIL  SrcCode/src/features/auth/components/ForgotPassword.empty.test.tsx > ForgotPassword validation > AUTH-FP-001: empty emai
l shows toast validation
ReferenceError: waitFor is not defined
 ❯ SrcCode/src/features/auth/components/ForgotPassword.empty.test.tsx:22:5
     20|
     21|     const rt = await import('react-toastify');
     22|     await waitFor(() => {
       |     ^
     23|       expect(rt.toast.error).toHaveBeenCalledWith('auth.validation.emailRequired');
     24|     });

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 Test Files  1 failed (1)
      Tests  1 failed (1)
   Start at  21:39:20
   Duration  1.18s (transform 79ms, setup 0ms, import 413ms, tests 33ms, environment 601ms)
```

AUTH-FP-002: ForgotPassword success -> sent=true

```
 ✓ SrcCode/src/features/auth/components/ForgotPassword.success.test.tsx > ForgotPassword success > AUTH-FP-002: successful rese
t sets sent=true and shows sent UI 44ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  21:40:38
   Duration  1.18s (transform 69ms, setup 0ms, import 409ms, tests 45ms, environment 592ms)
```

AUTH-FP-003: ForgotPassword auth/user-not-found -> toast đúng key lỗi

```
 ✓ SrcCode/src/features/auth/components/ForgotPassword.userNotFound.test.tsx > ForgotPassword error handling > AUTH-FP-003: sho
ws user-not-found toast when auth/user-not-found returned 43ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  21:41:33
   Duration  1.17s (transform 66ms, setup 0ms, import 395ms, tests 44ms, environment 601ms)
```

AUTH-OV-001: OTPVerification nhập đủ 6 số auto verify

```
 ✓ SrcCode/src/features/auth/components/OTPVerification.auto.test.tsx > OTPVerification auto-verify > AUTH-OV-001: typing 6 dig
its triggers auto verify and calls onVerificationSuccess 123ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  21:42:28
   Duration  1.09s (transform 60ms, setup 0ms, import 217ms, tests 124ms, environment 617ms)
```

AUTH-OV-002: OTPVerification verify fail thì clear input và focus ô đầu

```
 ✓ SrcCode/src/features/auth/components/OTPVerification.fail.test.tsx > OTPVerification failure behavior > AUTH-OV-002: on veri
fy fail, inputs cleared and focus set to first input 130ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  21:43:12
   Duration  1.10s (transform 53ms, setup 0ms, import 217ms, tests 131ms, environment 615ms)
```

AUTH-OV-003: OTPVerification resend success reset timer/canResend

```
 ✓ SrcCode/src/features/auth/components/OTPVerification.resend.test.tsx > OTPVerification resend behavior > AUTH-OV-003: on res
end success, reset timer and canResend; clear inputs and focus first 153ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  21:43:57
   Duration  1.09s (transform 52ms, setup 0ms, import 205ms, tests 154ms, environment 599ms)
```